

```
In [1]: import pandas as pd
import sqlite3

orders = pd.read_csv("store_orders.csv")
customers = pd.read_csv("store_customers.csv")

con = sqlite3.connect("store.db")

orders.to_sql("orders", con, if_exists="replace", index=False)
customers.to_sql("customers", con, if_exists="replace", index=False)
```

Out[1]: 40

```
In [2]: pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", con)
```

```
Out[2]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <thead> <th></th> <th>name</th> </thead> <tbody> <th>0</th>
<th>1</th> </tbody> <tbody></tbody>
```

orders

customers

```
In [3]: pd.read_sql("SELECT * FROM orders LIMIT 5;", con)
```

```
Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; }
</style> <thead> <th></th> <th>order_id</th> <th>customer_id</th>
<th>product</th> <th>category</th> <th>amount</th> <th>order_date</th>
</thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th>
</tbody> <tbody></tbody>
```

5000	4	paper	office	14.09	2026-05-25
------	---	-------	--------	-------	------------

5001	15	planner	office	51.86	2026-05-08
------	----	---------	--------	-------	------------

5002	34	shelf	furniture	116.25	2026-06-03
------	----	-------	-----------	--------	------------

5003	2	granola	snacks	19.72	2026-05-19
------	---	---------	--------	-------	------------

5004	17	monitor	electronics	284.80	2026-05-10
------	----	---------	-------------	--------	------------

```
In [4]: pd.read_sql("""
SELECT *
FROM orders
WHERE amount > 100
ORDER BY amount DESC;
""", con)
```

```
Out [4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
        .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;
        } </style> <thead> <th></th> <th>order_id</th> <th>customer_id</th>
        <th>product</th> <th>category</th> <th>amount</th> <th>order_date</th>
        </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th>
        <th>...</th> <th>79</th> <th>80</th> <th>81</th> <th>82</th> <th>83</th>
        </tbody> <tbody></tbody>
```

5153	20	dock	electronics	499.00	2026-05-25
5140	30	headset	electronics	295.36	2026-05-26
5047	901	headset	electronics	293.79	2026-05-15
5150	903	monitor	electronics	293.35	2026-06-06
5014	6	monitor	electronics	287.83	2026-05-22
...	...	...	...	...	...
5107	38	shelf	furniture	110.31	2026-05-17
5046	13	shelf	furniture	108.47	2026-05-28
5009	14	chair	furniture	107.20	2026-05-24
5007	36	dock	electronics	103.77	2026-05-21
5023	9	desk	furniture	100.27	2026-06-05

<p>84 rows × 6 columns</p>

```
In [5]: pd.read_sql("""
        SELECT category,
               COUNT(*) AS order_count,
               ROUND(AVG(amount), 2) AS avg_amount
        FROM orders
        GROUP BY category
        ORDER BY avg_amount DESC;
        """, con)
```

```
Out [5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
        .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;
        } </style> <thead> <th></th> <th>category</th> <th>order_count</th>
        <th>avg_amount</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
        <th>3</th> </tbody> <tbody></tbody>
```

electronics	60	190.57
-------------	----	--------

furniture	46	146.25
-----------	----	--------

office	51	30.13
--------	----	-------

snacks	43	14.62
--------	----	-------

```
In [6]: pd.read_sql("""
        SELECT o.order_id,
               o.customer_id,
               c.name,
               c.city,
               o.product,
               o.category,
               o.amount,
               o.order_date
        FROM orders AS o
        JOIN customers AS c
        ON o.customer_id = c.customer_id
        LIMIT 10;
        """, con)
```

```
Out [6]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;
} </style> <thead> <th></th> <th>order_id</th> <th>customer_id</th>
<th>name</th> <th>city</th> <th>product</th> <th>category</th>
<th>amount</th> <th>order_date</th> </thead> <tbody> <th>0</th> <th>1</th>
<th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th>
<th>8</th> <th>9</th> </tbody> <tbody></tbody>
```

5000	4	Dev Patel	Los Angeles	paper	office	14.09	2026-05-25
5001	15	Omar Garcia	Santa Monica	planner	office	51.86	2026-05-08
5002	34	Nina Patel	Venice	shelf	furniture	116.25	2026-06-03
5003	2	Ben Kim	Culver City	granola	snacks	19.72	2026-05-19
5004	17	Quinn Silva	Santa Monica	monitor	electronics	284.80	2026-05-10
5005	16	Priya Okafor	Culver City	monitor	electronics	66.10	2026-06-10
5006	27	Grace Silva	Culver City	headset	electronics	86.67	2026-06-08
5007	36	Priya Okafor	Culver City	dock	electronics	103.77	2026-05-21
5008	30	Jon Brooks	Los Angeles	headset	electronics	267.39	2026-06-08
5009	14	Nina Patel	Inglewood	chair	furniture	107.20	2026-05-24

```
In [7]: step5_sql = """
SELECT c.city,
       ROUND(SUM(o.amount), 2) AS total_spend
FROM orders AS o
JOIN customers AS c
ON o.customer_id = c.customer_id
GROUP BY c.city
ORDER BY total_spend DESC;
"""

step5_df = pd.read_sql(step5_sql, con)
step5_df
```

```
Out [7]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }
.dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;
} </style> <thead> <th></th> <th>city</th> <th>total_spend</th> </thead>
<tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th>
</tbody> <tbody></tbody>
```

Los Angeles	5862.99
Culver City	5316.88
Santa Monica	4017.93
Inglewood	2718.25
Torrance	1037.25
Venice	462.36

```
In [8]: print("Orders:", len(orders))

joined_rows = pd.read_sql("""
SELECT o.*
FROM orders AS o
JOIN customers AS c
ON o.customer_id = c.customer_id;
""", con)

print("Joined rows:", len(joined_rows))
```

Orders: 200

Joined rows: 195

<p>There are 200 orders in the original table, but the join has 195 rows. Five orders are missing because their customer IDs are not found in the customers table.</p>

<p>import matplotlib.pyplot as plt</p> <p>step5_df.plot(x="city",
y="total_spend", kind="bar", legend=False)</p> <p>plt.title("Total Spend by
City") plt.xlabel("City") plt.ylabel("Total Spend (\$)") plt.xticks(rotation=45)
plt.tight_layout() plt.show()</p>

<p>Business Question: Which product was ordered the most times?</p>

```
In [9]: pd.read_sql("""
SELECT product, COUNT(*) AS order_count
FROM orders
GROUP BY product
ORDER BY order_count DESC
LIMIT 1;
""", con)
```

```
Out [9]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; }  
        .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right;  
        } </style> <thead> <th></th> <th>product</th> <th>order_count</th> </thead>  
        <tbody> <th>0</th> </tbody> <tbody></tbody>  
        headset    23
```

<p>The headset was ordered the most. It appeared in 23 orders.</p>

<p>I would not include customer names when sharing the results. I would use
general or grouped data instead to keep customer information private.</p>

In []: